

MAHFUZUR RAHMAN

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EDUCATION

Bachelor of Science in Aeronautical Engineering (Major in Aerospace) May 2022 – June 2026
Military Institute of Science and Technology (MIST) Dhaka, Bangladesh
CGPA: **3.80 / 4.00** | Dean's List of Honour: 2023, 2024 & 2025
Relevant Coursework: Computational Fluid Dynamics, Aircraft Structures, Solid Mechanics, Thermodynamics, Heat Transfer, Advanced Materials Processing, Aerospace Propulsion, Numerical Analysis.

UNDERGRADUATE THESIS PROJECT

Thesis Title: Thermomechanical Behavior of Fiber-Reinforced Composite Materials
Supervisor: Dr. Shaikh Reaz Ahmed, Professor, Department of Mechanical Engineering, Bangladesh University of Engineering and Technology (BUET).
Synopsis:

- Developed a displacement-function-based finite difference numerical solver in MATLAB for coupled thermomechanical analysis of orthotropic composite structures.
- The solver resolves the complete thermoelastic field i.e. stress, displacement, and temperature distribution with accuracy close to ABAQUS FEM benchmarks under equivalent boundary conditions.

RESEARCH INTERESTS

- Thermomechanical Analysis of Composite and Metallic Structures.
- Computational Fluid Dynamics (CFD) and Numerical Modelling of aerospace flow fields.
- Structural Analysis and Design of Aerospace Structures.
- Fixed-Wing UAV Aerodynamic Design and Manufacturing.

ONGOING RESEARCH PROJECT

Research Title: Thermo-Mechanical Modeling of Generally In-Plane Anisotropic Fiber-Reinforced Composite Laminae by a Single Displacement Function
Supervisor: Dr. Shaikh Reaz Ahmed, Professor, Department of Mechanical Engineering, Bangladesh University of Engineering and Technology (BUET).
Synopsis:

- Extended the capability of the developed solver during undergrad thesis so that it can handle any thermal stress problem for any thin rectangular single layered anisotropic composite plate.
- The solver is equipped with unique ability to model mixed boundary conditions at any edge and output results resembling with real life conditions.

COMPETITION PROJECTS

AIAA 30th Annual Design/Build/Fly (DBF) 2026 – Team Lead Wichita, Kansas, USA
UAV: HeisenBird | Theme: Banner Towing Bush Plane

- Directed the full design/build/fly lifecycle of a 42-member team; led aerodynamic sizing in XFLR5 and AVL, structural design in SolidWorks
- Ranked 58th worldwide out of 175 competing universities; achieved 17th worldwide in Design Report (1st in Asia) and 18th worldwide in Proposal Report.

AIAA 29th Annual Design/Build/Fly (DBF) 2025 – Manufacturing Lead Tucson, Arizona, USA
UAV: Akashtori 2.0 | Theme: X-1 Supersonic Flight Program

- Managed composite manufacturing workflow including glass-fiber layup, rib fabrication, and structural assembly; ensured all airframe components met dimensional and weight budget requirements.
- Supported aerodynamic analysis in XFLR5 and AVL; performed structural assessments in ANSYS and ABAQUS for wing spar and fuselage frame under critical load cases.
- Ranked 39th worldwide out of 112 competing universities; secured 1st in Asia for Total Flight Mission Score.

INTERNSHIP EXPERIENCE

Industrial Trainee (May – June 2025)

Organization: **US-Bangla Airlines Ltd.**

CAMO & AMO departments, ATR 72-600 and Boeing 737-800 fleets. Reviewed Airworthiness Directives (ADs), MEL, MCC, and PPC; observed NDT procedures and SRM documentation; gained exposure to SMS processes within a Part-145 approved environment.

LEADERSHIP AND PROJECT MANAGEMENT

Department Representative (Jul 2025 – Jun 2026)	Organization: Department of Aeronautical Engineering, MIST Liaison between student body and faculty for academic scheduling, curriculum feedback, and co-curricular event coordination.
Vice President – Administration (Jun 2025 – Jun 2026)	Organization: MIST Aeronautics and Astronautics Club (MAAC)
Director of Event Management (Jun 2024 – Jun 2025)	Organization: MIST Drama and Film Society (MDFS)

ONLINE COURSES AND CERTIFICATIONS

MATLAB Onramp, Simulink Onramp

Organization: **MathWorks**

Python

Organization: **Kaggle**

AWARDS AND ACHIEVEMENTS

2026	17th out of 175 teams in Design Report in Design/Build/Fly (DBF) Competition	AIAA
2025	39th out of 112 teams in Design/Build/Fly (DBF) Competition	AIAA
2025	MIST Merit Scholarship - Spring 2025, for ranking 3rd in the class based on academic performance.	MIST
2023, 2024, 2025	MIST Dean's List of Honour , for earning a GPA greater than 3.75 in each academic year	MIST
2021	Board Scholarship (General) in HSC	Chattogram Education Board
2019	Board Scholarship (Talentpool) in SSC	Chattogram Education Board

SKILLS

Programming Language: **Python, MATLAB, C**

Design and Simulation: **SolidWorks, XFLR5, AVL, ANSYS, ABAQUS, Simulink**

Numerical Methods: **Finite Difference Method (FDM), Finite Element Method (FEM)**

Documentation: **LaTeX, Microsoft Office Suite, Mendeley**

LANGUAGES

Bangla: **Native**

English: **Fluent**

REFERENCES

Dr. Shaikh Reaz Ahmed

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Bangladesh University of Engineering and Technology (BUET)
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Saif Reza

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I hereby declare that all information provided above is true and accurate to the best of my knowledge.

Mahfuzur Rahman